

ADITYA ASUTOSH MISHRA

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github.com/adityaDev-2005

Summary

Aspiring Machine Learning Engineer and Full-Stack Developer with hands-on experience in designing and evaluating machine learning models across computer vision, natural language processing, and deep learning domains. Proficient in building convolutional neural networks (CNNs) for image classification tasks including deepfake detection, and developing backend services using Flask and FastAPI. Skilled in end-to-end ML pipelines encompassing data preprocessing, feature engineering, model training, and evaluation using Python, Scikit-learn, TensorFlow, and PyTorch. Passionate about developing intelligent, scalable systems that address real-world challenges in AI safety, vision, and language understanding.

Education

Vellore Institute of Technology

Bachelor of Technology in Information Technology

- CGPA: 8.61 / 10.0

Expected: July 2028

Vellore, India

Relevant Coursework

- Data Structures
- Machine Learning
- Linear Algebra
- Statistics

Technical Skills

Programming Languages: C, C++, Java, Python, JavaScript

Web Development & APIs: HTML, CSS, Flask, FastAPI, JSON

Databases: SQLite, PostgreSQL, SQL

Machine Learning & AI: NumPy, Pandas, Scikit-learn, TensorFlow, PyTorch, OpenCV, CNN, NLP, deep learning, neural networks

Data Processing: feature engineering, data preprocessing, data pipelines, Matplotlib, GeoPandas

Tools & Practices: Git, GitHub, version control, Jupyter Notebook

Projects

Deepfake Classification | *Python, TensorFlow, CNN, OpenCV, Scikit-learn*

College Hackathon

- Engineered a deep learning pipeline to classify real vs. deepfake images using Convolutional Neural Networks (CNNs), addressing the growing challenge of AI-generated media detection.
- Preprocessed and augmented image datasets using OpenCV and TensorFlow data pipelines to improve model generalization and reduce overfitting on unseen data.
- Designed and trained a CNN architecture with multiple convolutional and pooling layers, optimizing hyperparameters to maximize classification accuracy on validation data.
- Evaluated model performance using accuracy, precision, recall, and F1-score, achieving reliable detection of AI-manipulated media.
- **GitHub:** github.com/adityaDev-2005/model_arena_hack

Satellite Image Generator | *Python, GeoPandas, Shapely, Contextily, Matplotlib*

- Engineered a Python-based geospatial tool to generate high-resolution satellite images from location names or GPS coordinates, eliminating the need for specialized GIS software.
- Built end-to-end data pipelines using GeoPandas and Shapely for coordinate transformation, bounding box computation, and vector overlay rendering on raster tile maps.
- Integrated Contextily for dynamic web map tile fetching and Matplotlib for image export, supporting customizable zoom levels, resolutions, and multiple tile providers.
- **GitHub:** github.com/ToxicKoder/Satellite-image-generator

Drowsiness Detection System | *Python, OpenCV, dlib, Machine Learning*

- Architected a real-time computer vision system for driver drowsiness detection using facial landmark detection and the Eye Aspect Ratio (EAR) algorithm to classify eye closure states.
- Implemented optimized video stream processing with OpenCV achieving low-latency per-frame inference, enabling reliable real-time detection across varied lighting conditions.
- Deployed a threshold-based alert mechanism triggering audible warnings upon detecting prolonged eye closure, simulating a driver safety assist system.
- **GitHub:** github.com/ToxicKoder/Drowsiness-Detection-System